Pizza Sales Analysis - SQL Case Study

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Business Problem & Objective

- This project analyses pizza sales data using SQL to evaluate business performance, identify key revenue drivers, and uncover customer ordering trends.
- The goal is to provide actionable insights to support better decision-making.

Dataset Description

This dataset contains transactional pizza sales data, including order details, product categorie, sizes, and pricing. It is used to analyse revenue, sales trends, and customer ordering behaviour.

Total Revenue

```sql
SELECT
    ROUND(SUM(total_price), 2) AS Total_Revenue
FROM
    pizza_sales;
```

Result Grid

Total_Revenue
817860.05

Average Order Value

```sql
SELECT
    ROUND(SUM(total_price) /
    COUNT(DISTINCT order_id),2) AS Avg_Order_Value
FROM
    pizza_sales;
```

Result Grid

Avg_Order_Value
38.31

Total Number of Orders

```sql
SELECT
    COUNT(DISTINCT order_id) AS Total_Orders
FROM
    pizza_sales;
```

Result Grid

Total_Orders
21350

Total Pizzas Sold

```
SELECT
    SUM(quantity) AS Total_Pizzas_Sold
FROM
    pizza_sales;
```

Result Grid

	Total_Pizzas_Sold
▶	49574

Total Pizzas Sold by Pizza Category

```sql
SELECT
    pizza_category, SUM(quantity) AS Total_Pizzas_Sold
FROM
    pizza_sales
GROUP BY pizza_category
ORDER BY Total_Pizzas_Sold DESC;
```

Result Grid | Filter Rows:

pizza_category	Total_Pizzas_Sold
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Daily Order Trend

```sql
SELECT
    DAYNAME(order_date) AS Day_Name,
    COUNT(DISTINCT order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY Day_Name
ORDER BY Total_Orders DESC;
```

Result Grid | Filter Rows:

Day_Name	Total_Orders
Friday	3538
Thursday	3239
Saturday	3158
Wednesday	3024
Tuesday	2973
Monday	2794
Sunday	2624

Monthly Order Trend

```
SELECT
    MONTHNAME(order_date) AS Month_Name,
    COUNT(DISTINCT order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY Month_Name
ORDER BY Total_Orders DESC;
```

Result Grid | Filter Rows:

Month_Name	Total_Orders
July	1935
May	1853
January	1845
August	1841
March	1840
April	1799
November	1792
June	1773
February	1685
December	1680
September	1661
October	1646

Percentage Contribution of each Category to Total Revenue

```sql
SELECT
    pizza_category,
    ROUND(SUM(total_price) , 2) AS Total_Sales,
    Concat(ROUND(SUM(total_price) * 100 /
    sum(sum(total_price)) over(), 2),'%')
    AS Percentage_of_Sales
FROM
    pizza_sales
GROUP BY pizza_category;
```

Result Grid | Filter Rows:

pizza_category	Total_Sales	Percentage_of_Sales
Classic	220053.10	26.91%
Veggie	193690.45	23.68%
Supreme	208197.00	25.46%
Chicken	195919.50	23.96%

Total 5 Pizzas by Revenue

```sql
SELECT
    pizza_name,
    ROUND(SUM(total_price), 2) AS Total_Revenue
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768.00
The California Chicken Pizza	41409.50
The Classic Deluxe Pizza	38180.50
The Spicy Italian Pizza	34831.25

Total 5 Pizzas by Quantity

```
SELECT
    pizza_name, SUM(quantity) AS Total_Quantity
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Total 5 Pizzas by Total Orders

```
SELECT
    pizza_name,
    Count(Distinct order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders Desc
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Orders
The Classic Deluxe Pizza	2329
The Hawaiian Pizza	2280
The Pepperoni Pizza	2278
The Barbecue Chicken Pizza	2273
The Thai Chicken Pizza	2225

Bottom 5 Pizzas by Revenue

```sql
SELECT
    pizza_name, SUM(total_price) AS Total_Revenue
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue ASC
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Revenue
The Brie Carre Pizza	11588.50
The Green Garden Pizza	13955.75
The Spinach Supreme Pizza	15277.75
The Mediterranean Pizza	15360.50
The Spinach Pesto Pizza	15596.00

Bottom 5 Pizzas by Quantity

```sql
SELECT
    pizza_name, SUM(quantity) AS Total_Quantity
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity Asc
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Quantity
The Brie Carre Pizza	490
The Mediterranean Pizza	934
The Calabrese Pizza	937
The Spinach Supreme Pizza	950
The Soppressata Pizza	961

Bottom 5 Pizzas by Total Orders

```sql
SELECT
    pizza_name,
    Count(Distinct order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders Asc
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Orders
The Brie Carre Pizza	480
The Mediterranean Pizza	912
The Calabrese Pizza	918
The Spinach Supreme Pizza	918
The Chicken Pesto Pizza	938

Top 5 Orders with above Average Spending

```sql
with order_totals as (
      select order_id, sum(total_price) as order_value
      from pizza_sales
      group by order_id)
select order_id,
       order_value
from order_totals
where order_value > (
      select avg(order_value) from order_totals)
order by order_value DESC
Limit 5;
```

Result Grid | Filter Rows:

order_id	order_value
18845	444.20
10760	417.15
1096	285.15
6169	284.00
740	280.95

Total 5 Pizzas by Total Orders

```
SELECT
    pizza_name,
    Count(Distinct order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders Desc
LIMIT 5;
```

Result Grid | Filter Rows:

pizza_name	Total_Orders
The Classic Deluxe Pizza	2329
The Hawaiian Pizza	2280
The Pepperoni Pizza	2278
The Barbecue Chicken Pizza	2273
The Thai Chicken Pizza	2225

Most Popular Pizza Size per Category

```sql
SELECT *
FROM (
    SELECT pizza_category,
           pizza_size,
           SUM(quantity) AS total_qty,
           dense_rank() OVER(PARTITION BY pizza_category
           ORDER BY SUM(quantity) DESC) AS rnk
    FROM pizza_sales
    GROUP BY pizza_category, pizza_size
) t
WHERE rnk = 1;
```

Result Grid | Filter Rows:

pizza_category	pizza_size	total_qty	rnk
Chicken	L	4932	1
Classic	S	6139	1
Supreme	L	4564	1
Veggie	L	5403	1

KEY INSIGHTS

1. OVERALL BUSINESS PERFORMANCE

The business generated 800k revenue from 21k+ orders, showing consistent demand.

2. The Classic category generates the hightest revenue (220k), making it the top-performing segment.

3. Large-sized pizzas dominate sales across categories, indicating preference for group or value meals.

KEY INSIGHTS

4.Friday records the highest number of orders, showing strong end-of-week demand.

5.Pizzas like Brie Carre generate very low revenue (~11k), indicating poor performance.

Business Recommendations

1. Increase promotions for top-performing pizzas like Classic category and Thai Chicken Pizza, as they contribute the highest revenue.
2. Review low-performing items such as Brie Carre pizza, which contribute minimal revenue.Consider menu redesign, price adjustment, or removal to improve operational efficiency.
3. Introduce weekend combo deals to capitalise on peak demand.

CONCLUSION

The analysis identified key revenue drivers, customer demand patterns, and product performance trends.

High performing categories like Classic pizzas and large sizes contribute significantly to overall sales.

Sales patterns show strong demand during weekends and peak periods.

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This project demonstrates my ability to transform raw data into actionable insights to support data-driven business decisions.